INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M13 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M13 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Indian Head Division, Naval Surface Warfare Center (CAGE Codes 19200 and 19203)

10520974	Thruster, Cartridge Actuated, M13 Metal Parts Assembly
8836128	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M13 or M19
8836129	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M13 or M19
9217388	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M13 or M19

U.S. Air Force

TPO 00-756-3501	Thruster, Cartridge Actuated, M13

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

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3.4 Hydrostatic pressure. The body shall withstand a hydrostatic pressure of 11,500 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Torque. Torque requirements shall be as specified in table I. No movement or deformation shall occur upon application of the breakaway torque.

TABLE I. Torque requirements.

Component	A	B	C
	Run on (minimum)	Seating or Bottoming, Including A (inch pounds)	Breakaway (inch pounds)
Piston and piston fastener	—	150 ± 20	75 minimum

3.5.1 Piston rotation. Upon completion of assembly, the piston fastener shall be free to rotate 360 degrees in a clockwise direction.

3.6 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.6.

3.10 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as

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specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

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4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Pin, Shear, Firing Pin, Drawing 10520486, covering a detail of Drawing 10520974.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Surface finish improper	Visual

4.5.2.2 Pin, Shear, Piston, Drawing 10520487, covering a detail of Drawing 10520974.

Critical		
None defined		
Major		
101	Diameter, max	Gage
102	Total length, min	Gage

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103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Surface finish improper	Visual

4.5.2.3 Ring, Drawing 10520489, covering a detail of Drawing 10520974.

Critical
None defined

Major		
101	Concentricity of inside diameter with outside diameter	Gage
102	Concentricity of pin holes	Gage
103	True position of pin holes	Gage
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
201	Outside diameter	Gage
202	Inside diameter	Gage
203	Diameter of pin holes	Gage
204	Depth of pin holes	Gage
205	Width	Gage
206	Location of pin holes	Gage
207	Marking missing, incorrect, or illegible	Visual
208	Radius or chamfer incorrect	Gage
209	Surface finish improper	Visual

4.5.2.4 Body, Drawing 10520976, covering a detail of Drawing 10520974.

Critical
None defined

Major		
101	Pitch diameter of large thread, max	Gage
102	Minor diameter of large thread, max	Gage
103	Pitch diameter of thread, inlet port, max	Gage

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104	Minor diameter of thread, inlet port, max	Gage
105	Pitch diameter of thread, by-pass port, max	Gage
106	Minor diameter of thread, by-pass port, max	Gage
107	Concentricity of large cavity with long cavity	Gage
108	Concentricity of the small end of the counterbore with bore	Gage
109	Concentricity of lugs	Gage
110	Concentricity of bore with long cavity	Gage
111	Concentricity of grooves with bore	Gage
112	Concentricity of pin holes	Gage
113	True position of pin holes	Gage
114	True position of lugs	Gage
115	Diameter of long cavity, max	Gage
116	Diameter of bore, max	Gage
117	Diameter of lugs, min	Gage
118	Diameter of pin holes	Gage
119	Diameter of cavity, inlet port	Gage
120	Diameter of grooves	Gage
121	Depth to bottom of large cavity, max	Gage
122	Depth of counterbore, inlet port	Gage
123	Depth of counterbore, by-pass port	Gage
124	Depth of effective thread, inlet port, min	Gage
125	Depth of effective thread, by-pass port, min	Gage
126	Depth to large cavity	Gage
127	Length of thread, inlet port, max	Gage
128	Length of thread, by-pass port, max	Gage
129	Location of lugs	Gage
130	Location of inlet port	Gage
131	Angle of chamfer, inlet port	Gage
132	Angle of chamfer, by-pass port	Gage
133	Angular location of hole, inlet port	Gage
134	Diameter of chamfer, inlet port	Gage
135	Diameter of chamfer, by-pass port	Gage
136	Diameter of relief undercut	Gage
137	Diameter of undercut, long cavity, if applicable, max	Gage
138	Profile of grooves	Gage
139	Internal radii forward end long cavity	Gage
140	Radii at base of lugs	Gage
141	Thread damaged or not full	Visual
142	Metal defective	Visual
143	Protective finish inadequate or defective	Visual
144	Surface finish of large cavity, long cavity and angles in gas ports, improper	Visual
145	Burr or sharp corners	Visual
146	Foreign matter	Visual
Minor		
201	Pitch diameter of large thread, min	Gage
202	Pitch diameter of thread, inlet port, min	Gage
203	Pitch diameter of thread, by-pass port, min	Gage

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204	Pitch diameter of setscrew thread	Gage
205	Minor diameter of setscrew thread, max	Gage
206	Diameter of long cavity, min	Gage
207	Diameter of large cavity	Gage
208	Diameter of lugs, max	Gage
209	Diameter of bore, min	Gage
210	Diameter of undercut, large thread, max	Gage
211	Diameter of counterbore, small end	Gage
212	Diameter of counterbore, inlet port	Gage
213	Diameter of counterbore, by-pass port	Gage
214	Diameter of hole, inlet port	Gage
215	Diameter of hole, by-pass port	Gage
216	Diameter of body, small end	Gage
217	Depth to bottom of relief undercut	Gage
218	Depth to bottom of large cavity, min	Gage
219	Depth to counterbore, small end	Gage
220	Depth of setscrew thread, max	Gage
221	Depth to long cavity	Gage
222	Length to intersection of face, inlet port, and head diameter	Gage
223	Total length	Gage
224	Length of lugs	Gage
225	Length of effective setscrew thread, min	Gage
226	Length of head	Gage
227	Length from lugs to face at small end	Gage
228	Length of by-pass port	Gage
229	Width across head, at lug	Gage
230	Width of relief undercut	Gage
231	Location of hole, inlet port	Gage
232	Location of pin holes	Gage
233	Location of grooves	Gage
234	Location of by-pass port	Gage
235	Centrality of head	Gage
236	Height of by-pass port	Gage
237	Depth of cavity, inlet port	Gage
238	Width of grooves	Gage
239	Width of undercut at large cavity	Gage
240	Spot face improper	Visual
241	Marking missing, incorrect, or illegible	Visual
242	Radius or chamfer incorrect, except as otherwise classified	Gage
243	Surface finish improper, except as otherwise classified	Visual

4.5.2.5 Breech, Drawing 10520977, covering a detail of Drawing 10520974.

Critical		
1	Surface finish of cavity improper	Gage
Major		
101	Concentricity of basic diameter of boss with body	Gage
102	Concentricity of body with cavity	Gage

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103	Concentricity of striker hole with cavity	Gage
104	Concentricity of boss cavity with cavity	Gage
105	Concentricity of groove with body	Gage
106	True position of pin hole	Gage
107	Perpendicularity of boss face with body	Gage
108	Diameter of cavity, max	Gage
109	Diameter of body, min	Gage
110	Diameter of groove	Gage
111	Diameter of pin hole, max	Gage
112	Height of boss	Gage
113	Profile of groove	Gage
114	Escapement hole missing	Visual
115	Metal defective	Visual
116	Protective finish inadequate or defective	Visual
117	Burr or sharp corner	Visual
118	Foreign matter	Visual

Minor

201	Diameter of striker shoulder cavity	Gage
202	Diameter of cavity, min	Gage
203	Diameter of body, max	Gage
204	Diameter of pin hole, min	Gage
205	Diameter of striker hole	Gage
206	Diameter of boss	Gage
207	Diameter of relief groove	Gage
208	Diameter of holes in relief groove	Gage
209	Depth of striker hole	Gage
210	Length from bottom of cavity to boss face	Gage
211	Total length	Gage
212	Location of basic diameter	Gage
213	Location of pin hole	Gage
214	Location of holes in relief groove	Gage
215	Diameter of escapement hole	Gage
216	Width of groove	Gage
217	Width of flange	Gage
218	Location of groove	Gage
219	Location of relief groove	Gage
220	Location of escapement hole	Gage
221	Countersink, escapement hole, missing	Visual
222	Marking missing, incorrect, or illegible	Visual
223	Radius or chamfer incorrect	Gage
224	Surface finish improper, except as otherwise classified	Visual

4.5.2.6 Pin, Firing, Drawing 10520978, covering a detail of Drawing 10520974.

Critical
None defined

Major

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101	Concentricity of groove with body	Gage
102	Concentricity of striker with body	Gage
103	Concentricity of striker shoulder with body	Gage
104	True position of pin hole	Gage
105	Diameter of groove	Gage
106	Diameter of body, min	Gage
107	Diameter of striker	Gage
108	Diameter of pin hole, max	Gage
109	Profile of striker	Gage
110	Profile of groove	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish of tip improper	Visual
114	Burr or sharp corner	Visual
115	Foreign matter	Visual

Minor

201	Diameter of striker shoulder	Gage
202	Diameter of body, max	Gage
203	Diameter of pin hole, min	Gage
204	Total length	Gage
205	Length of striker	Gage
206	Length from body to tip of striker	Gage
207	Location of pin hole	Gage
208	Depth of pin hole	Gage
209	Depth of slot in base	Gage
210	Width of groove	Gage
211	Width of slot in base	Gage
212	Location of groove	Gage
213	Marking missing, incorrect, or illegible	Visual
214	Radius or chamfer incorrect	Gage
215	Surface finish improper, except as otherwise classified	Visual

4.5.2.7 Cap, Base, Drawing 10520982, covering a detail of Drawing 10520975.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Diameter of thread lead	Gage
104	Thread damaged or not full	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr or sharp corners	Visual

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108	Foreign matter	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of pilot	Gage
203	Diameter of wrench holes	Gage
204	Depth of wrench holes	Gage
205	Total length	Gage
206	Length of thread lead	Gage
207	Length of body	Gage
208	Location of wrench holes	Gage
209	Marking missing, incorrect, or illegible	Visual
210	Radius or chamfer incorrect	Gage
211	Surface finish improper	Visual

4.5.2.8 Fastener, Piston, Drawing 10520983, covering a detail of Drawing 10520980.

Critical
None defined

Major		
101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Minor diameter of internal thread, max	Gage
105	Concentricity of pitch diameter of external thread with pitch diameter of internal thread	Gage
106	Concentricity of pilot with internal thread	Gage
107	Threads damaged or not full	Visual
108	Metal defective	Visual
109	Protective finish inadequate or defective	Visual
110	Burr	Visual
111	Foreign matter	Visual
Minor		
201	Pitch diameter of external thread, max	Gage
202	Pitch diameter of internal thread, min	Gage
203	Diameter of pilot	Gage
204	Diameter of flange	Gage
205	Diameter of undercut, external	Gage
206	Diameter of undercut, internal	Gage
207	Depth to bottom of undercut, threaded cavity	Gage
208	Length of pilot	Gage
209	Total length	Gage
210	Length to flange	Gage
211	Width of undercut, external	Gage
212	Width of undercut, internal	Gage
213	Width across flats	Gage

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214	Location of flats	Gage
215	Marking missing, incorrect, or illegible	Visual
216	Radius or chamfer incorrect	Gage
217	Surface finish improper	Visual

4.5.2.9 Piston, Drawing 10520984, covering a detail of Drawing 10520979.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of thread with body	Gage
104	Concentricity of shoulder with body	Gage
105	Concentricity of head with body	Gage
106	Concentricity of groove with head	Gage
107	Concentricity of cavity with body	Gage
108	Diameter of head, min	Gage
109	Diameter of body, min	Gage
110	Diameter of shoulder, min	Gage
111	Diameter of groove	Gage
112	Profile of groove	Gage
113	Thread damaged or not full	Visual
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Surface finish of body improper	Visual
117	Burr or sharp corners	Visual
118	Foreign matter	Visual

Minor

201	Pitch diameter of thread, max	Gage
202	Diameter of head, max	Gage
203	Diameter of body, max	Gage
204	Diameter of shoulder, max	Gage
205	Diameter of undercut, head, if applicable	Gage
206	Diameter of cavity	Gage
207	Diameter of undercut at shoulder	Gage
208	Length to shoulder	Gage
209	Total length	Gage
210	Length of shoulder	Gage
211	Location of head	Gage
212	Location of groove	Gage
213	Location of holes in body	Gage
214	Diameter of escapement hole in head	Gage
215	Diameter of holes in body	Gage
216	Depth of cavity	Gage
217	Width of groove	Gage
218	Width of undercut, body, if applicable	Gage

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219	Width of thread undercut	Gage
220	Marking missing, incorrect, or illegible	Visual
221	Radius or chamfer incorrect	Gage
222	Surface finish improper, except as otherwise classified	Visual

4.5.2.10 Base, Cap and Pellet Assembly, Drawing 10520975, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Protrusion of pellets	Gage
102	Pellet missing, loose, or improperly inserted	Manual/Visual
103	Thread damaged	Visual
104	Burr	Visual
105	Foreign matter	Visual

Minor

201	Location of pellets	Gage

4.5.2.11 Piston and Pellet Assembly, Drawing 10520979, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Protrusion of pellet	Gage
102	Pellet missing, loose, or improperly inserted	Manual/Visual
103	Thread damaged	Visual
104	Burr	Visual
105	Foreign matter	Visual

Minor

201	Location of pellets	Gage

4.5.2.12 Piston Fastener and Pellet Assembly, Drawing 10520980, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Protrusion of pellet	Gage
102	Pellet missing, loose, or improperly inserted	Manual/Visual
103	Thread damaged	Visual
104	Burr	Visual
105	Foreign matter	Visual

Minor

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201	Location of pellets	Gage

4.5.2.13 Breech and Firing Pin, Subassembly, consisting of Breech, Drawing 10520977, Firing Pin, Drawing 10520978, and O-ring, MS29513-9, covering a detail of Drawing 10520974.

Critical		
1	Firing pin protrusion	Gage
Major		
101	Protective finish inadequate or defective	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
None defined		

4.5.2.14 Metal Parts Assembly, Drawing 10520974.

Critical		
None defined		
Major		
101	Piston not free to rotate 360 degrees clockwise	Manual
102	Any part missing, except as otherwise classified	Visual
103	Improper assembly	Visual
104	Thread damaged or not full	Visual
105	Foreign matter	Visual
Minor		
201	Protective finish inadequate or defective	Visual
202	Barrier material missing from setscrews	Visual
203	Protective cap or plug missing	Visual
204	Pellet missing from piston fastener	Visual

4.5.2.15 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8836128.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.16 Box, Packing, Ammunition, Drawing 8836129.

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Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.17 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217388.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each body shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Breakaway torque. Each piston fastener and piston shall be subjected to a breakaway torque test. Failure to comply with the requirements of column C of table I shall be cause for rejection of the unit.

4.5.3.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.6 shall be cause for rejection of the lot.

4.5.3.5 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

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4.5.3.6 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.6. Failure of any unit in the inspection sample to comply with 3.9 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish each body shall be mounted in the testing device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Breakaway torque. A torque wrench, set at the torque requirement specified in table I, shall be used in conjunction with a fixture to determine the adequacy of the joint. The torque shall be applied to the piston fastener.

4.6.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.5 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.6 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8836128 and 8836129. Level B shall be in accordance with Drawings 8836128 and 9217388.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-756-3501.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

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Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E11)

Review activity:
Air Force - 70

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